

The field, the rail and the ledger

An audit of Wottle's match experience as it is built today, and a proposal to rebuild the game panel around three objects and nothing else.

Read from source: components/match/*, components/game/*, app/styles/board.css, lib/constants/*, docs/prd_and_requirements/wottle_game_rules.md, docs/superpowers/specs (0-59, 0-71), ds-bundle render sheets.

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01 WHAT THE GAME IS

Sixty seconds, one swap, a hundred letters

Wottle asks one thing of a player every round: find, among a hundred letters, the single swap that makes the most valuable word, and make it before the other player does. Everything else on the screen exists either to support that act or to record its result. That is the test every element has to pass.

Reading the rules document rather than the PRD (the PRD still describes a 5-minute chess clock and eight scoring directions; the engine runs ten rounds of 60 seconds on two axes), the game rests on three skills and two resources.

SKILL · SCANNING Word-search perception. The eye runs rows and columns looking for near-	SKILL · VALUATION Letter points (Krafla table, P 7, X 10) plus (length-2)×5 plus a combo bonus.	SKILL · TIMING The first submission wins coordinate conflicts and freezes tiles instantly, denying them to the
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words. The field must be a clean text to read, not a set of objects to inspect.	Valuation happens after a candidate is found, never while scanning.	opponent. Speed is rewarded; the clock is a weapon, not just a limit.
RESOURCE · TIME 60 s per round, both clocks run at once, yours pauses when you submit. After submitting, the only time that matters is the opponent's.	RESOURCE · TERRITORY Scored letters freeze in the scorer's colour. Territory shrinks the field, feeds the tiebreak, and, by rule \$3.5a, always comes in whole words of three or more.	RHYTHM · A ROUND Set → think → commit → reveal → settle. Five beats, ten times. The panel's job is to make each beat unmistakable with exactly one signal.

Principles derived from the rules

1. **The field is the game.** Nothing covers it, nothing competes with it, and nothing on it that is not a letter or a state of a letter.
2. **Show words, not tiles.** Territory is stored per tile but earned per word. Draw the word.
3. **One clock, one question.** Before you submit: how long do I have? After: how long until the reveal? Never two answers at once.
4. **One ledger.** Every number about the match (scores, rounds, words, territory) lives in one scorecard, in one place, in one register.
5. **Colour is identity.** Two inks belong to the two players. The system itself speaks only in ink and paper; alarm is weight and motion, not a third hue.
6. **Each beat has one signal.** A round announcement, a lock banner, a ring flash, a popup and a list update for the same event is five signals for one fact.
7. **Teach once, then leave.** Rules are for the first match and a help key, not for the panel of every match.

Fourteen objects around a hundred letters

The desktop match screen, recreated from `MatchClient.tsx`, `board.css` and the rail card components, at the moment Birna (player A) has picked her first tile in round 4 and Kári (player B) has already played. Values, spacing and colours are the production tokens.

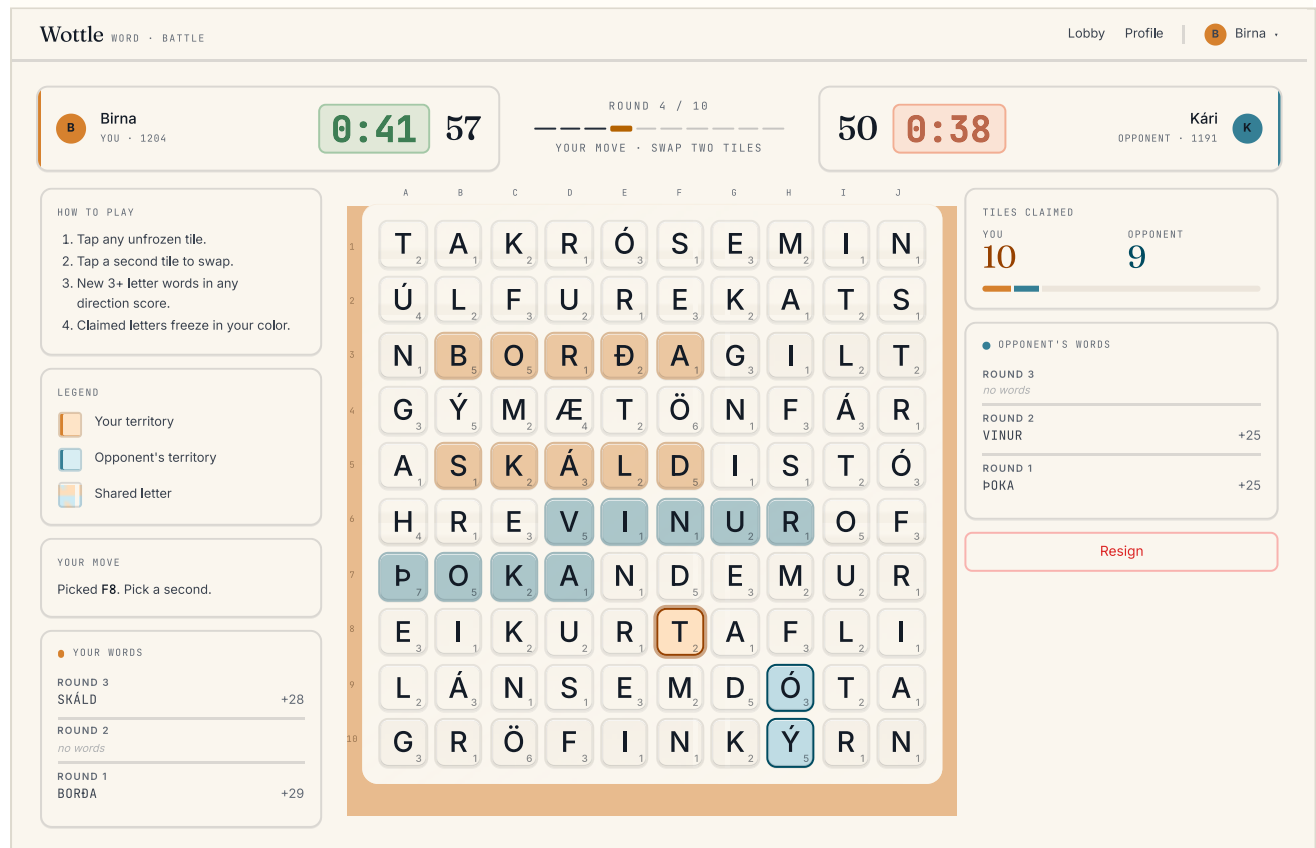


Fig. 1 – Desktop match screen as built ($\geq 900\text{px}$). `TopBar.tsx` · `HudCard.tsx` · `MatchCenterChrome.tsx` · `RoundPipBar.tsx` · `MatchLeftRail.tsx` (`HowToPlayCard`, `LegendCard`, `YourMoveCard`) · `ScoredWordsCard.tsx` ×2 · `TilesClaimedCard.tsx` · `BoardCoordLabels.tsx` · `BoardGrid.tsx`.

Census of the panel

Persistent objects around the field (top bar, 2 HUD cards, centre chrome, 7 rail cards, resign)	14
...of which change state during a round (two clocks, "Your move")	3
Large numerals in the HUD strip (two 36px clocks, two 34px scores)	4
Places the round number is shown (label, pip bar, "R4" on each mobile bar)	3

Glyphs on the field that are not letters (letter values ×100, coordinate labels ×20)	120
Distinct signals fired by one round resolution (FLIP, ring flash, held ring, delta popup, card update ×3, announce overlay)	8
Meanings carried by the colour ochre (brand, player A, current pip, waiting clock, lock banner, waiting frame)	6

The panel is not ugly. It is a well-made magazine page with a puzzle in the middle. The trouble is that the puzzle is the game, and the page keeps talking.

03

FINDINGS

Nineteen findings, four of them bugs

Each finding names the file it was read from. Tags: **BUG** is wrong today, **NOISE** competes with the field, **REDUNDANT** repeats a fact already shown, **RISK** costs the player a round or a match.

F1 **NOISE**

A · FIELD

A hundred letter-value numerals sit in the scanning field

Every tile prints its Krafla value at 0.35em and 50% opacity. Valuation happens after a candidate is found, not while scanning; during the scan these are one hundred extra glyphs the eye must skip.

BoardGrid.tsx board-grid__tile-score · board.css .board-grid__tile-score

F2 NOISE

A · FIELD

Letterpress tiles spend contrast on decoration

Each of 100 cells carries a two-stop gradient, four layered shadows, a 0.65rem radius and a 2px hover lift; the board wrapper adds a gradient and a tinted drop shadow. It is the Scrabble-tile trope, and it lowers letter contrast against the cell.

`board.css` `.board-grid__cell`, `.board-grid`

F3 REDUNDANT

A · FIELD

Coordinate labels serve a card, not the player

A–J and 1–10 exist so the Your-move card can say “Picked F8”. Nothing in play uses chess notation: no notation input, no replay, no chat. They add 20 glyphs and 18px to the field's footprint.

`BoardCoordLabels.tsx` · `YourMoveCard.tsx` `formatCoord`

F4 NOISE

A · FIELD

The lock banner covers the first row of letters

“Move submitted — waiting for opponent” is absolutely positioned 0.5rem from the top of the grid and sits over row 1, at exactly the moment a player who has finished wants to pre-scan for the next round.

`board.css` `.board-grid__lock-banner` · `BoardGrid.tsx` `showLockBanner`

F5 BUG

A · FIELD

The round announcement is invisible

`.round-announce` sets `color: var(--paper)` on a 92% paper background. Only the text-shadow glow renders. “Round 5” and “Final Round” have not been readable since the theme flipped from dark to paper.

`board.css` `.round-announce` (`color`, `background`)

F6 NOISE

B · CHROME

Seven cards frame the board, two of them are a tutorial

How to play and Legend are shown in every match to every player forever. Your move, Your words, Opponent's words and Tiles claimed are four containers for facts that belong in one scorecard.

MatchLeftRail.tsx · MatchClient.tsx rails

F7 BUG

B · CHROME

"Hidden from opponent until both submit" is no longer true

Since #210 the swap is broadcast and animated on the opponent's board the moment it is submitted (rules §2). The Your-move card still tells the player the opposite.

YourMoveCard.tsx SubmittedState · wottle_game_rules.md §2

F8 REDUNDANT

B · CHROME

Two clocks that say the same thing

Both clocks count the same 60 seconds. The opponent's clock carries one bit of information (submitted or not) yet is rendered as a 36px bold numeral co-equal with the score. Four large numbers compete in one strip.

HudCard.tsx · board.css .hud-card__clock (0-59)

F9 REDUNDANT

B · CHROME

The round number is shown three times

"Round 4 / 10" in text, a ten-pip bar beneath it, and on mobile "R4" on both player bars.

MatchCenterChrome.tsx · RoundPipBar.tsx · PlayerPanel.tsx

F10 RISK

B · CHROME

Navigation stays sticky and the height budget is wrong

The TopBar (wordmark, Lobby, Profile, user menu) remains pinned during a live match. board.css budgets 48px of chrome for the board's height, but the top bar, HUD strip and paddings spend about 190px, so on displays under ~820px tall the bottom rows scroll off.

app/layout.tsx · TopBar.tsx · board.css .match-layout__board --chrome-height

F11 RISK

C · STATE & COLOUR

Picking and having played look identical

PLAYER_A_SELECTED_BG equals PLAYER_A_LOCKED_BG and the borders match, by design. The most important state distinction in the round, “I am choosing” versus “I have committed”, has no visual difference on the tile.

lib/constants/playerColors.ts · board.css --selected / --locked

F12 BUG

C · STATE & COLOUR

Errors flash in the opponent's colour

.board-grid__cell--invalid borders in --p2 (blue), which is player B's identity colour, not --bad. For player B an invalid move looks like their own selection; the PRD asked for a red flash.

board.css .board-grid__cell--invalid · wottle_prd.md §7.2

F13 NOISE

C · STATE & COLOUR

Ochre means six things

Brand accent, player A identity (--p1 ≈ --ochre), the current pip, the waiting clock (--warn is a near hue), the lock banner and the pulsing waiting frame. A player cannot learn what ochre means because it means everything.

globals.css --ochre/--p1/--warn · board.css

F14 NOISE

C · STATE & COLOUR

One clock, five tones, two renderings

active, warning (yellow→red), critical (blink), expired, waiting: rendered as 16% tints on desktop and 85% saturated fills on mobile, so the same state looks different on the two surfaces. The submitted glow hard-codes rgba(245,158,11), Tailwind amber, outside the token set.

HudCard.tsx · TimerDisplay.tsx · deriveClockUrgency.ts

F15 NOISE

C · STATE & COLOUR

Territory is painted per tile, so words disappear

A frozen five-letter word is five separately tinted squares; a crossing is a diagonal split. The rules guarantee every frozen letter belongs to a run of three or more (§3.5a), yet the field never shows the run.

BoardGrid.tsx FROZEN_COLORS · playerColors.ts BOTH_GRADIENT

F16 NOISE

D · RHYTHM

The reveal, the round's climax, is eight small signals

FLIP swap, 450ms delay, 700ms ring flash, a held 3px ring, an 11px green “+18 letters, +10 length” for 3s, three cards updating, and a 1.2s announce. The highest-drama moment has the smallest type on the page and no single beat.

MatchClient.tsx round-recap effect · ScoreDeltaPopup.tsx · board.css keyframes

F17 RISK

D · RHYTHM

The second tap is irrevocable

handleTileClick submits on the second tap; the swap is broadcast and may be instantly scored server-side, so there is no undo. A mis-tap costs the whole round. The PRD's desktop drag-to-swap was never built.

BoardGrid.tsx handleTileClick → animateSwap · wottle_prd.md §1.3

F18 BUG

D · RHYTHM

Copy describes a different game

Matchmaking reads “Ranked · 5+0 · Icelandic nouns”; the lobby hero says “outrun the chess clock”. The match is ten 60-second rounds validated against the full BÍN inflected list.

MatchmakingClient.tsx · LobbyHero.tsx · game-config.ts · rules §3.3

Warm Editorial is the period style, not Wottle's

Cream paper, Fraunces italic display, mono uppercase kickers, hairline cards with left-stripe accents and Inter body is the prevailing 2024–26 editorial-app look. It reads as a magazine; nothing in it comes from Icelandic letterforms or from word-search.

ds-bundle/README.md · globals.css · HudCard left stripe

04

PROPOSAL

Three objects: field, rail, ledger

Everything the panel currently does is re-homed into three objects. The **field** is the hundred letters and the state of each letter. The **rail** is a single line of time along the field's top edge. The **ledger** is one scorecard that holds every number: players, totals, round, words, territory. There are no cards, no top bar, no overlays and no third colour.

Field	Rail	Ledger
REPLACES	REPLACES	REPLACES
Board grid, letterpress tiles, coordinate labels, letter-value numerals, lock banner, round-announce overlay, legend card	Both HUD clocks, both mobile timer pills, five clock tones, the pulsing "waiting" frame	Two HUD cards, centre chrome, pip bar, "Your move", "Your words", "Opponent's words", "Tiles claimed", score-delta popup, history modal, resign button, top bar

Identity: set in type, marked in two inks

Warm Editorial borrows its look from magazines. The proposal takes its look from the game's own material: a printed word-search page, set in a slab serif so Þ, Ð, Æ and Ö carry weight, marked up by two players with two inks. Paper and ink are the system; coral and teal are the players. No gradients, no shadows, no rounded cards, no italic display type.

PALETTE · FIVE VALUES, NO MORE



paper ink rule coral · teal ·
#FFFDF7 #0F1A24 #E6E2D6 A B
#E4573D #147D7A

Player inks are used at full strength for letters and at 16% as the tint of a settled word. There is no alarm colour: urgency is thickness, size and blink in ink. Coral and teal are separable for the common colour-vision deficiencies and also differ in lightness, so runs read apart in greyscale.

TYPE · ZILLA SLAB FOR LETTERS, RED
HAT MONO FOR NUMBERS

ÞÐÆÖÝ
SKÁLD KÝR

57 · R4 · 50 +18
0:09

A slab's even stroke and flat serifs keep a hundred capitals aligned like set type, and its Icelandic glyphs are drawn, not borrowed. The mono carries every numeral so digits line up in the ledger. Body copy outside the match (lobby, profile) can stay in a text face; the match itself uses only these two.

The match screen, rebuilt

The same moment as Fig. 1: round 4, 41 seconds left, Birna has picked T, Kári has played. Fourteen objects have become three.

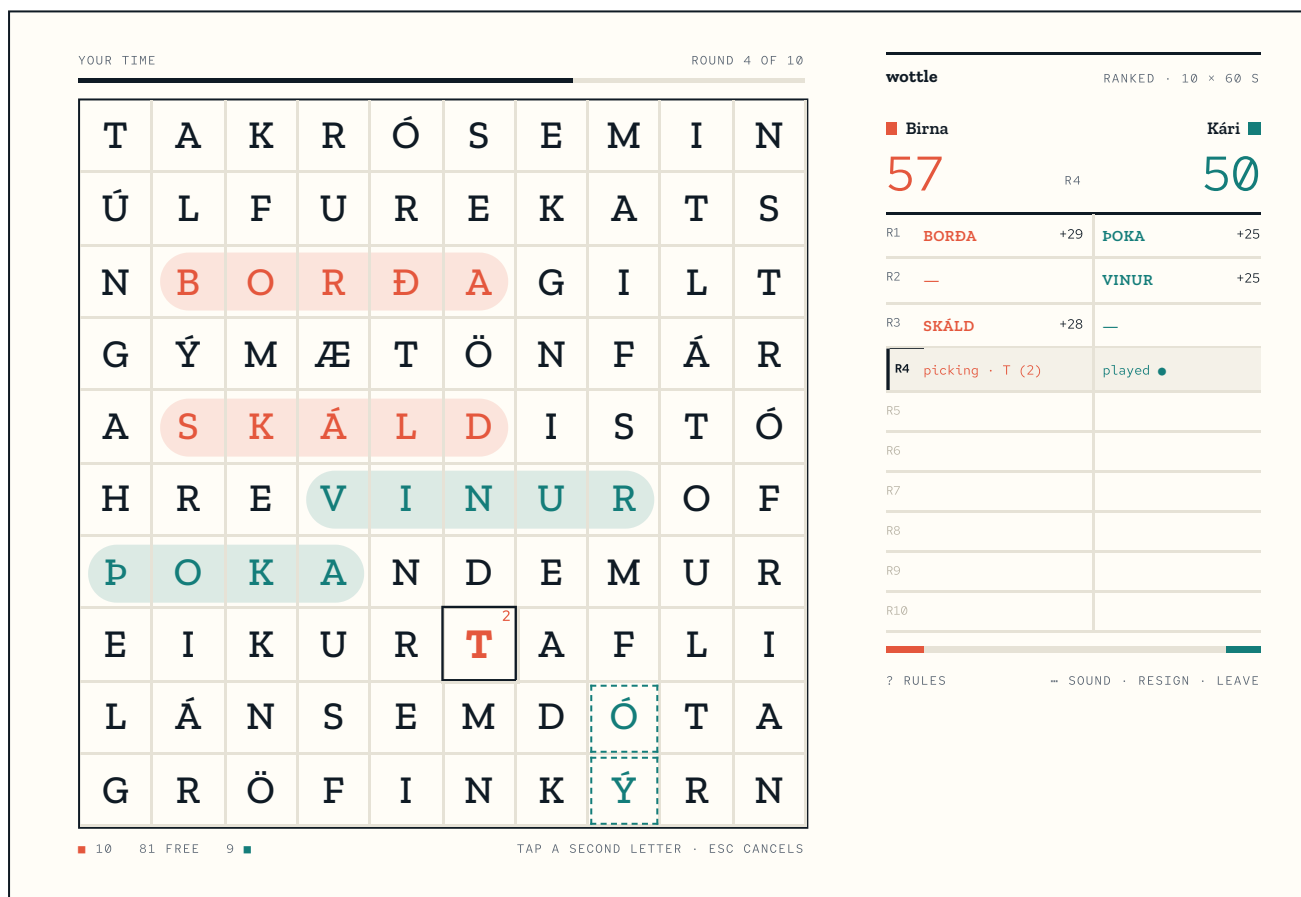


Fig. 2 – Proposed desktop match screen at 1180px. Field 660px, ledger 340px, no top bar. Settled words are drawn as runs; Kári's played letters are pinned in teal; Birna's pick is ringed in ink with its value shown.

The field

A ruled grid of one hundred capitals in ink on paper. Cells are flat; the rules between them are a hairline; the frame is 1.5px ink. Letters take about 55% of the cell height, larger than today's 5cqi, because the numerals that shared the cell are gone. Eight things can happen to a letter, and each has one mark.

A	S	K	Á	L
T	Ö	Ý	E	N
O	R	R	Ð	U

A	N	D	E	M
U	R	T ²	A	F
S	E	M	D	Ó

Settled words are runs

One capsule per scored word, tinted in the scorer's ink, letters set in that ink. A crossing shows both runs; the shared letter is set in ink black. Territory reads as words, which is what it is. Partial freezes (the 24-tile floor, rule §6) draw the run only over the frozen letters.



The rail

One line along the top edge of the field, as wide as the field, depleting left to right over 60 s. It sits in peripheral vision, so the eye never leaves the letters to read a clock. After you play it switches to the opponent's remaining time, drawn dashed. Under 15 s it doubles in thickness, grows a numeral and blinks at 1 Hz; under reduced motion it holds solid. This keeps O-59's intent (a prominent, urgent clock) without a second hue or a second clock.

The ledger

A scorecard with ten rows, one per round, and two columns, one per player. It is the only place a number appears. The header carries names, inks and totals; the live row shows each player's state in the round (*picking*, *played*, then the words); the foot carries territory and the two quiet controls (rules, menu). Because the ledger is a table and not a stack of cards, it also serves as round history, post-game recap and spectator view without a second component.

- The score delta is not a popup. At the reveal, the word and its points are written into the live row while the total in the header counts up over 400 ms. The eye is already there.

Picked, previewed, pinned

A picked letter turns your ink, grows 8% and gets an ink square ring; its value appears in the corner, the only time a value is shown. Touching a second letter previews the exchange in place. Once committed, both letters are *pinned*: your ink, dashed ring, no fill. The opponent's played letters pin in their ink the moment they play, as rule §2 broadcasts them.

- Combo and duplicate outcomes appear as suffixes in the row: +18 · +2 combo, SKÁL · again · 0.
- Territory is a two-ink bar under the table (10 · 81 free · 9). The tiebreak is visible all match without a card.
- Elo, mode and format belong to the pre-match screen and the ledger's caption line, not the HUD.
- Hover or tap a row and the field highlights those runs; the same gesture drives the post-game recap.

05
ROUND CHOREOGRAPHY

Five beats, one signal each

Nothing is ever placed over the field: a player who has finished thinking is already scanning for the next round, and the current lock banner and round announcement take that time away.
Detail of rows 7–10, columns D–J.

1 Set

0.4 s

A	N	D	E	M	U	R
U	R	T	A	F	L	I
S	E	M	D	Ó	T	A
F	I	N	K	Ý	R	N

R4 · Birna – · Kári –

The rail refills, a new ledger row opens with the round number. The field is untouched; whoever pre-scanned during the last reveal keeps that lead.

2 Think

≤ 60 s

A	N	D	E	M	U	R
U	R	T	A	F	L	I
S	E	M	D	Ó	T	A
F	I	N	K	Ý	R	N

R4 · picking T (2) · played ●

Birna has picked T; its value shows. Kári has played: his two letters pin in teal the instant the broadcast lands, and the live row says played.

3 Commit

one tap

A	N	D	E	M	U	R
U	R	T	A	F	L	I
S	E	M	D	Ó	T	A
F	I	N	K	Ý	R	N

R4 · played · played ●

Birna previews R↔T, taps again. Both letters pin in coral; the rail turns dashed and now shows the opponent's time (here none is left: both have played).

4 Reveal

1.2 s

A	N	D	E	M	U	R
U	R	T	A	F	L	I
S	E	M	D	Ó	T	A
F	I	N	K	Ý	R	N

R4 · TAFL +18 · KÝR +13

Runs draw themselves along each word, 400 ms per word, staggered. The ledger writes the word and points into the row; totals count 57→75 and 50→63.

5 Settle

0.6 s

A	N	D	E	M	U	R
U	R	T	A	F	L	I
S	E	M	D	Ó	T	A
F	I	N	K	Ý	R	N

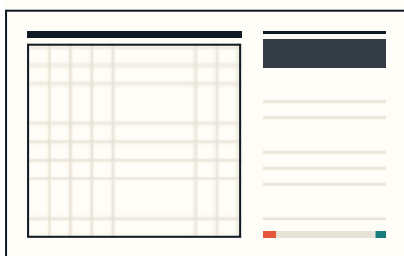
R5 · Birna – · Kári –

Pins fade, live runs settle to the territory tint, the territory bar updates, and R5 opens. Nothing has covered the field at any point.

Fig. 3 – One round from Birna's seat. Rail, field detail and the ledger's live row at each beat. All motion is 400 ms or shorter and disappears under prefers-reduced-motion, where runs and pins appear in their end state.

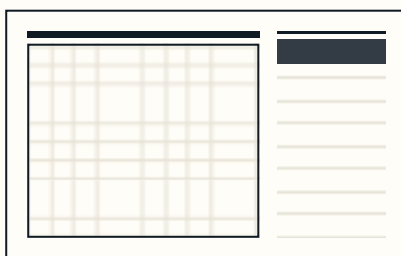
The field always fits without scrolling

The layout rule is a single constraint: the field is the largest square that fits the viewport after the rail and a one-row ledger header are placed. On wide screens the full ledger sits beside the field; on narrow screens it collapses to its header and live row, and the rest pulls up. There is no third breakpoint and no sticky navigation during a match.



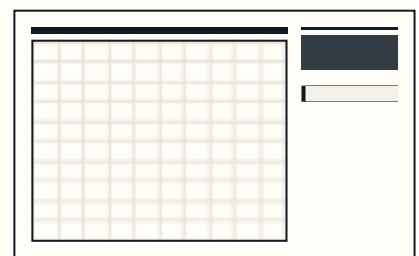
Desktop ≥ 1100

Field left, ledger 340px right, 56px gutter. The field is $\min(\text{viewport height} - 120\text{px}, 720\text{px})$ square, so tiles run 50–70px.



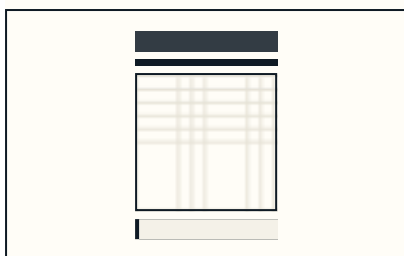
Laptop 900–1100

Same arrangement; the ledger narrows to 260px and drops word points to a hover. The field keeps priority: on a 13-inch display at 100% zoom it is 560px, never below the fold. Today's layout budgets 48px of chrome (board.css) but spends about 190px.



Phone landscape

Field fills the height; the ledger shows header and live row only, with the history behind a tap. Rail sits above the field as always.



Phone portrait

Header row, rail, full-width field, live row. One clock, one score pair, one round number:

the two compact bars with two clocks, two scores and two "R4" labels become one row.



Fig. 4 – Phone as built (PlayerPanel.tsx x2, TimerDisplay.tsx, BoardCoordLabels, History trigger).



Fig. 5 – Phone proposed. Header row, rail, field, live row. The field is 30px wider than today because the coordinate labels and card borders are gone; tiles are 37px with a 44px hit area.

Pick, preview, commit

Today the second tap submits an irrevocable move for a 60-second round. The proposal inserts one beat: the second tap *preview*s the exchange in place, and the player commits by tapping either previewed letter again (or Enter). The preview lets the player read the word they are about to make, which is the check most mis-taps fail. The cost is one tap; the alternative is a lost round. An "instant commit" setting can remove the beat for players who want it.

tap A	Pick. Letter turns your ink, grows 8%, ink square ring, value shown. Sound: tile-select.
tap B	Preview. The two letters exchange places in 150 ms; both carry a dotted ink ring. Nothing is sent. Any new run through A or B is <i>not</i> hinted; that would be a spoiler.
tap A or B again · Enter	Commit. Rings turn dashed (pinned), rail switches to the opponent's time, ledger row writes <i>played</i> . Sound: valid-swap, haptic.
tap elsewhere · Esc · tap A	Cancel. Preview reverses; pick clears. No sound.
drag A → B (pointer)	Same as tap A, tap B: a preview, not a commit. Fulfils PRD \$1.3 desktop drag without a separate path.
tap a frozen or pinned letter	Letter shakes 300 ms in its own ink; the live row reads <i>frozen</i> · <i>Kári R2</i> for 2 s. No red, no blue: an error is a fact about the board, not a colour.
long-press · hover 600 ms	Show the letter's value in the corner. Values never show otherwise; the full Krafla table lives behind the ledger's ?.
opponent plays	Their two letters pin in their ink at once (rule §2). If one of them was your pick, your pick clears and the live row says so, as spec 042 FR-004 already requires.
keyboard	Arrows move a focus ring in ink, Space picks or previews, Enter commits. Coordinates stay in the accessible name of each cell for screen readers; they are no longer printed.

first match

Three sentences before round 1, over a live field, dismissed by the first pick: *Swap two letters. Words of three or more score and freeze in your ink. Sixty seconds, ten rounds.* Never shown again; ? in the ledger foot reopens it.

08

MIGRATION

Fix, remove, replace, then restyle

The order matters: the first two steps ship inside the current look and each removes something the third step would otherwise have to migrate. No server or type changes anywhere; every step is client-side and covered by the existing Playwright flow.

P0 · fix

F5 round-announce colour; F7 "hidden from opponent" copy; F12 invalid flash to a non-player colour; F18 matchmaking and hero copy. Half a day. `board.css` · `YourMoveCard.tsx` · `MatchmakingClient.tsx` · `LobbyHero.tsx`

P1 ·
remove

Letter-value numerals (show on pick only), coordinate labels (keep in aria), How-to-play and Legend cards (first-match overlay), Your-move card, the lock banner and the pulsing frame, "R4" on the mobile bars, the top bar during a match. One to two days. `BoardGrid.tsx` · `BoardCoordLabels.tsx` · `MatchLeftRail.tsx` · `PlayerPanel.tsx` · `app/layout.tsx`

P2 ·
ledger

New MatchLedger fed by `accumulatedWords`, `accumulatedScores`, `frozenTiles` and the two timers, all of which MatchClient already holds. Retire HudCard, MatchCenterChrome, RoundPipBar, ScoredWordsCard, TilesClaimedCard, ScoreDeltaPopup and the history modal. Rail replaces both clock renderings and reuses `deriveClockUrgency` for its thresholds. Three days.

P3 ·
field

Runs layer derived from `wordHistory.coordinates` and `frozenTiles` (clip to frozen letters for partial freezes); pick / preview / commit state machine in BoardGrid; reveal choreography replacing the recap phase machine's ring-flash, held-ring and announce timers. Three to four days, plus the Playwright two-player flow update.

**P4 ·
identity**

Tokens (five values), Zilla Slab and Red Hat Mono via next/font, and the pre-match and post-game screens rebuilt on the ledger. Lobby and profile can follow later; they are not in the loop.

Open questions for the team

- O-59 asked for prominent timers and got two 36px clocks. The rail is more prominent by area and less by numerals. Is a numeral-free clock above 15 s acceptable to the people who filed O-59?
- The commit beat adds roughly 300 ms to every move in a game where the first submission wins conflicts. Ship it default-on with an instant-commit setting, or default-off?
- Partial freezes can leave a run half frozen. Draw the run over frozen letters only (proposed), or over the whole word with unfrozen letters hollow?
- Coral/teal are proposed inks; the brand ochre can stay as player A if the brand needs it, provided it stops meaning anything else.

Board letters and words from the ds-bundle canonical fixture; scores computed with the Krafla table and $(n-2) \times 5$.

Figures 1 and 4 are recreations of production components, not screenshots.